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Chapter 5, The Bayesian Linear Model

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Assessing Model Quality

- ▶ Motivation: there is nothing in the previous discussions that would *per se* prevent one from producing incorrect and overly sensitive posterior distributions for the unknown parameters of interest that are:
 - ▶ overly sensitive to the form of the prior,
 - ▶ a poor fit to the data,
 - ▶ or generally fragile WRT given assumptions.

Assessing Model Quality (cont.)

- ▶ Three general approaches:
 - ▶ **Posterior Predictive Checks:** formally or informally comparing model predictions with observed outcomes.
 - ▶ **Sensitivity Analysis:** the *informal* process of altering assumptions according to researcher intuition with the objective of determining the extent to which these changes modify the posterior distribution (global and local).
 - ▶ **Global Robustness Evaluation:** *systematic* analysis of the degree to which posterior inferences are affected by both potential misspecification of the prior and influential data points.
 - ▶ **Local Robustness Evaluation:** using differential calculus to determine the volatility of specific reported results.

Likelihood Function Robustness

- ▶ The likelihood function component of the model is no less suspect than the prior.
- ▶ Very little concern in all statistical literatures about likelihood robustness.
- ▶ One method (Shyamalkumar 2000) replace the specified likelihood function with a more “uncertain” one and compare posterior quantiles (i.e. normal to Cauchy).
- ▶ Another method (Lavine 1991) is to estimate the posterior replacing the likelihood function with a nonparametric form that resembles the likelihood function but follows the data more closely.

Predictive Checks

- ▶ Consider a model of support for abortion under different scenarios using survey data from Britain in consecutive years from 1983 to 1986.
- ▶ The panel data for 264 respondents is collected annually by McGrath and Waterton (1986) where seven scenarios are provided and these respondents have the option of expressing support or disagreement for abortion.
- ▶ The full collection of these seven queries do not fall into an obvious ordinal scale, so we will treat them here as nominal and judge total support for abortion as a binomial test for each respondent at each wave of the panel
- ▶ The scenarios are: (1) the woman decides on her own that she does not wish to have the child, (2) the couple agree that they do not wish to have the child, (3) the woman is not married and does not wish to marry the man, (4) the couple cannot afford any more children, (5) there a strong chance that the baby has a biological defect, (6) the woman's health is seriously endangered by the pregnancy, and (7) the woman became pregnant as a result of rape.
- ▶ So each respondent can produce an outcome from 0 to 7.

Predictive Checks

- ▶ The model is specified for $i = 1, \dots, 264$ respondents across $j = 1, \dots, 4$ panel waves:

$$y_{ij} \sim \mathcal{BN}(n, p_{ij})$$

$$\text{logit}(p_{ij}) = \beta_{0,j} + \beta_{1,i} X_{1,i}$$

$$\beta_{0,i} \sim \mathcal{N}(\mu_0, \tau_0)$$

$$\beta_{1,i} \sim \mathcal{N}(\mu_1, \tau_1)$$

$$\mu_0 \sim \mathcal{N}(0, 100)$$

$$\mu_1 \sim \mathcal{N}(0, 100)$$

$$\tau_0 \propto \mathcal{C}_{half}(25)$$

$$\tau_1 \propto \mathcal{C}_{half}(25)$$

- ▶ where the second term in the normals is a variance and half-Cauchy priors are positive-support, zero-centralized forms with a scale term equal to $A = 25$, $f(\tau) = (1 + \tau/A)^{-1}$, $\tau > 0$.
- ▶ Here $X_{1,i}$ is the i th person's self-identified religion: Catholic (1), Protestant (2), Other (3), and No Religion (4), and $n = 7$ at each wave for each person.
- ▶ The Gibbs sampler is run for 20,000 iterations dispensing with the first 10,000 values.

Global Sensitivity Analysis

- ▶ A broad approach that evaluates a wide range of: alternative prior specifications, forms of the link function , missing data implications, error sensitivity.
- ▶ The purpose of global sensitivity is to vary the *widest* possible range of assumptions
- ▶ Leamer's original objective was to determine the maximum amount that the model assumptions can be varied without dramatically changing posterior inferences (1985, p. 308).
- ▶ The idea was to use a broad class of neighboring assumptions.

Problems with Global Sensitivity Analysis

- ▶ It is not always clear what qualifies as a [neighborhood prior](#), and what range of these alternative priors constitutes sufficient testing criteria.
- ▶ Sometimes global sensitivity analysis is just not possible due to “an inability to sufficiently refine the usually subjective inputs of model, prior, and loss” (Berger 1986).
- ▶ There can therefore be an enormous number and variety of specifications to consider and evaluate with difficult inclusion criteria.
- ▶ Alternative: ‘backward’ sensitivity analysis or [prior partitioning](#) as advocated by Carlin and Louis (1996). The idea is to fix the posterior at some interesting or realistic configuration and see what types of priors are consistent with reaching this prior given the data.

Local Sensitivity Analysis

- ▶ The more modest and realizable process of making minor changes in the prior parameterization while looking at the subsequent posterior effects.
- ▶ This has the advantage of realizing many of the benefits of global sensitivity analysis such as indicating posterior fragility, but at a much lower cost in terms of effort and reporting.
- ▶ Generally modifying the parameters of the prior is done to produce a more diffuse form of the prior than that used in the initial analysis.

Local Sensitivity Analysis (cont.)

- Change m in the normal mean:

$$\pi(\mu|\mathbf{X}, \sigma^2) \sim \left[\left(\frac{m}{s^2} + \frac{n\bar{x}}{\sigma_0^2} \right) / \left(\frac{1}{s^2} + \frac{n}{\sigma_0^2} \right), \left(\frac{1}{s^2} + \frac{n}{\sigma_0^2} \right)^{-1} \right]$$

- Normal variance: change the s^2 prior parameter by setting it to double or half its value, $s_1^{2*} = s^2/2$, $s_2^{2*} = 2s^2$ and observe the effect on the resulting posterior.
- Here m , can be analyzed by adding and subtracting one prior standard deviation, $m_1^* = m - s$, $m_2^* = m + s$, and recalculating the posterior distribution two more times using m_1^* and m_2^* separately as substitutes for m .

Local Sensitivity Analysis (cont.)

- ▶ Exponential/gamma model, ad hoc changes: $\beta = 10, 4, 2, 1$ and $\alpha = 10, 17.5, 30, 42.5$ (fixing the non-changed value at the midpoint).

Local Sensitivity Analysis (cont.)

- ▶ Exponential/gamma model, 50% prior variance (α/β^2) change.

- ▶ $\alpha = 5, 10, 15$ and $\beta = 1.414, 2, 2.449$.

Poisson-Gamma Model for the Log of the Recidivism Rate

- ▶ Data from: <http://www.doc.state.ok.us/CHARTS/recidivi.htm#by%20crime%20type>
(unknown offenses at release time excluded).
- ▶ Four specification for modeling the log of returns:
 - ▷ three with differently parameterized gamma priors producing the posterior:
$$\pi(\lambda|\mathbf{X}) \propto \lambda^{(\alpha + \sum x_i) - 1} \exp[-\lambda(\beta + n)],$$
 - ▷ and one with a bounded uniform prior producing the posterior: $\pi(\lambda|\mathbf{X}) \propto \lambda^{\sum x_i} \exp[-\lambda n]$.

Global Robustness

- ▶ Analyze insensitivity to misspecification of the prior *and* insensitivity to influential datapoints through the likelihood function (the latter is technically “resistance”).
- ▶ The primary objective of Bayesian robustness is to determine the *posterior* sensitivity to substantively reasonable changes in model assumptions and thereby understand the context of uncertainty introduced by the prior, the likelihood function, and if appropriate, the loss function.
- ▶ We can see how much the bounds of the 95% HPD region change under different prior assumptions. Other statistics of obvious interest include the posterior mean, various quantiles, and measures of dispersion.

Global Robustness with ϵ -Contaminated Priors

- ▶ Berger (1984) points out that no single prior distribution can be shown to be absolutely correct and a “class” of priors therefore provides more information about model uncertainty.
- ▶ Starting with a specified prior, $p_0(\theta)$, identify a wider class of **contamination** forms that include this prior but also broadens the range of specific forms considerably: $\mathcal{Q}$ containing alternatives $q(\theta)$.
- ▶ It is also necessary to also give a subjective probability indicating how *uncertain* we are about the selection of $p_0(\theta)$, label this probability ϵ .

Global Robustness with ϵ -Contaminated Priors

- ▶ The class priors evaluation is the subset of Q given by all forms of $p(\theta)$ that satisfy:

$$\Gamma(Q, \epsilon) = \{p(\theta) : p(\theta) = (1 - \epsilon)p_0(\theta) + (\epsilon)q(\theta), q(\theta) \in Q\}.$$

So $\Gamma(Q, \epsilon)$ is an ϵ -weighted compromise between the initial prior and the selected overarching class of alternatives.

- ▶ Suggestions for defining Q :
 - ▶ all possible contaminants,
 - ▶ all symmetric and unimodal forms,
 - ▶ unimodality-preserving forms,
 - ▶ normals.

Global Robustness with ϵ -Contaminated Priors (cont.)

- ▶ Moreno (2000) gives the following example: suppose the data consist of a single data point distributed $\mathcal{N}(\theta, 1)$, the prior is specified as $\mathcal{N}(0, 2)$, and we set $\epsilon = 0.2$
- ▶ Consider three contaminants of the form:
 - ▶ all distributions ($q(\theta) \in Q_A$),
 - ▶ all distributions with the same median as the prior ($q(\theta) \in Q_M$),
 - ▶ all distributions with the same quartiles as the prior ($q(\theta) \in Q_4$).
- ▶ Tradeoff: a very broad Q may lead to a large range and a very narrowly defined Q may lead to a small range (generalizability), but neither one of these results tells as much as a broad class leading to a narrow range or a narrow class leading to a broad range.

Global Robustness with ϵ -Contaminated Priors (cont.)

- The quantity of interest is the 95% HPD interval for θ , and the table shows the most extreme upper and lower bounds reached over each contaminating class for five different x :

x	Q_A		Q_M		Q_4	
0.5	0.82	0.96	0.84	0.96	0.91	0.96
1.0	0.77	0.97	0.82	0.96	0.87	0.96
2.5	0.71	0.97	0.79	0.96	0.86	0.96
3.0	0.36	0.98	0.52	0.97	0.66	0.97
4.0	0.13	0.99	0.20	0.99	0.50	0.97

- Generally the more we restrict the contamination class, the smaller the posterior range.
- Since the data value is always greater than the prior mean here, the infimum of the posterior HPD intervals is the more variable component.
- The data matters in all of this: the range increases as the data point moves away from the prior mean.

Global Robustness with ϵ -Contaminated Priors (cont.)

- ▶ Problem 1: the specification of the contamination class may be difficult in cases where the obvious choices are not reasonable.
- ▶ Problem 2: the calculations of posterior bounds can be foreboding in high-dimension problems.
- ▶ Problem 3: the results may be inconclusive.

Local Robustness

- ▶ Because the scope of doing global sensitivity in a *completely* rigorous and thorough manner is beyond the resources of many research projects, there is a motivation for a more globally limited, but still informative, paradigm for assessing robustness in Bayesian models.
- ▶ Define the locally perturbed prior to be a weighting of the original prior and the perturbation:

$$p^*(\theta) = (1 - \epsilon)p(\theta) + (\epsilon)q(\theta),$$

where $q(\theta)$ is a single identified disturbance prior here, and ϵ is not an expression of doubt about the original prior but a weighting of how much to perturb with $q(\theta)$.

Local Robustness (cont.)

- Given $L(\theta|\mathbf{x})$, the subsequent ϵ -weighted posterior is:

$$\pi_{\epsilon}(\theta|\mathbf{x}) = \frac{(1 - \epsilon)p(\theta)L(\theta|\mathbf{x}) + (\epsilon)q(\theta)L(\theta|\mathbf{x})}{(1 - \epsilon)m_p(\mathbf{x}) + (\epsilon)m_q(\mathbf{x})}$$

now weighted by the marginal posterior densities of the data using the original prior ($m_p(\mathbf{x})$) and the disturbance prior ($m_q(\mathbf{x})$):

$$m_p(\mathbf{x}) = \int_{\Theta} p(\theta)L(\theta|\mathbf{x})d\theta, \quad m_q(\mathbf{x}) = \int_{\Theta} q(\theta)L(\theta|\mathbf{x})d\theta.$$

- Further define the difference between the ϵ -weighted posterior and the nondisturbed posterior as ϵ goes to zero:

$$\begin{aligned} D(q) &= \lim_{\epsilon \rightarrow 0} (\pi_{\epsilon}(\theta|\mathbf{x}) - p(\theta)L(\theta|\mathbf{x})/m_p(\mathbf{x})) \\ &= \left(\frac{q(\theta)L(\theta|\mathbf{x})}{m_q(\mathbf{x})} - \frac{p(\theta)L(\theta|\mathbf{x})}{m_p(\mathbf{x})} \right) \frac{m_q(\mathbf{x})}{m_p(\mathbf{x})}, \end{aligned}$$

which gives $\|D(q)\|$ as the local (infinitesimal) posterior sensitivity from disturbing $p(\theta)$ by $q(\theta)$,

can be thought of as the derivative of the posterior with prior $p(\theta)$ in the direction of $q(\theta)$.

Posterior Predictive Distribution

- *Prior predictive distribution* of a new data value, x_{new} before observing the full dataset:

$$p(x_{new}) = \int_{\Theta} p(x_{new}, \theta) d\theta = \int_{\Theta} p(x_{new}|\theta)p(\theta) d\theta.$$

i.e. the marginal distribution of an unobserved data value is the product of the prior for θ and the single variable PDF or PMF, integrating out this parameter.

Posterior Predictive Distribution (cont.)

- Now consider the distribution of a new data point, x_{new} *after* the full iid data set, $\mathbf{X}$, has been observed:

$$\begin{aligned} p(x_{new}|\mathbf{x}) &= \int_{\Theta} p(x_{new}, \theta|\mathbf{x}) d\theta \\ &= \int_{\Theta} \frac{p(x_{new}, \theta|\mathbf{x})}{p(\theta|\mathbf{x})} p(\theta|\mathbf{x}) d\theta \\ &= \int_{\Theta} p(x_{new}|\theta, \mathbf{x}) p(\theta|\mathbf{x}) d\theta. \end{aligned}$$

This can be simplified since x_{new} and $\mathbf{x}$ are assumed independent given θ :

$$= \int_{\Theta} p(x_{new}|\theta) p(\theta|\mathbf{x}) d\theta.$$

Posterior Predictive Distribution (cont.)

- ▶ Normal example: $X_1, X_2, \dots, X_n$ are distributed iid $\mathcal{N}(\mu, \sigma_0^2)$, where σ_0^2 is known but μ is unknown. Place a normal prior on μ according to $\mu \sim \mathcal{N}(m, s^2)$.

- ▶ Posterior for μ :

$$\pi(\mu|\mathbf{x}) \propto \exp \left[-\frac{1}{2} \left(\frac{1}{s^2} + \frac{n}{\sigma_0^2} \right) \left(\mu - \frac{\left(\frac{m}{s^2} + \frac{n\bar{x}}{\sigma_0^2} \right)}{\left(\frac{1}{s^2} + \frac{n}{\sigma_0^2} \right)} \right)^2 \right].$$

- ▶ Re-expressed in terms of its mean and variance:

$$\pi(\mu|\mathbf{x}) \propto \exp \left[-\frac{1}{2\sigma_1^2} (\mu - \mu_1)^2 \right]$$

$$\text{where } \sigma_1^2 = \left(\frac{1}{s^2} + \frac{n}{\sigma_0^2} \right)^{-1} \quad \mu_1 = \frac{\left(\frac{m}{s^2} + \frac{n\bar{x}}{\sigma_0^2} \right)}{\left(\frac{1}{s^2} + \frac{n}{\sigma_0^2} \right)}.$$

Posterior Predictive Distribution (cont.)

- So the posterior predictive distribution is: bgm

$$\begin{aligned} p(x_{new}|\mathbf{x}) &= \int_{\mu} p(x_{new}|\mu)p(\mu|\mathbf{x})d\mu \\ &\propto \int_{\mu} \exp\left[-\frac{1}{2}\left(\frac{(x_{new}-\mu)^2}{\sigma_0^2} + \frac{(\mu-\mu_1)^2}{\sigma_1^2}\right)\right] d\mu. \end{aligned}$$

- Summary statistics from this PPD:

$$\begin{aligned} E[x_{new}|\mathbf{x}] &= E[E(x_{new}|\mu, \mathbf{x})|\mathbf{x}] \\ &= E[E(\mu|\mathbf{x})] \\ &= E[\mu] = \mu_1. \end{aligned}$$

$$\begin{aligned} \text{Var}[x_{new}|\mathbf{x}] &= E[\text{Var}(x_{new}|\mu, \mathbf{x})|\mathbf{x}] + \text{Var}[E(x_{new}|\mu, \mathbf{x})|\mathbf{x}] \\ &= E[\sigma_0^2|\mathbf{x}] + \text{Var}[\mu|\mathbf{x}] \\ &= \sigma_0^2 + \sigma_1^2/n. \end{aligned}$$

Model Selection

- ▶ Suppose we are choosing between: $M_1, M_2, \dots, M_K$.
- ▶ For each κ , determine a model prior: $p(M_\kappa)$.
- ▶ Recall that the “integrated likelihood” is:

$$p(\mathbf{x}|M_\kappa) = \int \underbrace{p(\boldsymbol{\theta}_\kappa|M_\kappa, \mathbf{x})p(\boldsymbol{\theta}_\kappa|M_\kappa)}_{\text{likelihood} \times \text{prior}} d\boldsymbol{\theta}_\kappa.$$

Model Selection, Cont.

- This lets calculate the *posterior model probability* for each model, $\kappa = 1 \dots K$:

$$\pi(M_\kappa | \mathbf{x}) = \frac{p(\mathbf{x} | M_\kappa) p(M_\kappa)}{\sum_{j=1}^K p(\mathbf{x} | M_j) p(M_j)}.$$

- Note that $P(M_j)$ are values not distributions here (typically summing to one).

Simple Bayesian Model Averaging

- ▶ Use a simple averaging scheme from Raftery with a linear regression specification.
- ▶ Start with the posterior mean and variance for the j^{th} coefficient of the κ^{th} model, $\boldsymbol{\theta}_j(\kappa)$ and $Var_{\boldsymbol{\theta}_j}(\kappa)$.
- ▶ Consider:

$$P(\boldsymbol{\theta}_j \neq 0|\mathbf{x}) = \sum_{\boldsymbol{\theta}_j \in M_\kappa} \pi(M_\kappa|\mathbf{x})$$

which is just the “posterior probability that $\boldsymbol{\theta}_j$ is in the model,” as well as:

$$E[\boldsymbol{\theta}_j|\mathbf{x}] \approx \sum_{i=1}^K \hat{\boldsymbol{\theta}}_j(\kappa) \pi(M_\kappa|\mathbf{x})$$

and:

$$Var[\boldsymbol{\theta}_j|\mathbf{x}] \approx \sum_{\kappa=1}^K \left[(Var_{\boldsymbol{\theta}_j}(\kappa) + \hat{\boldsymbol{\theta}}_j(\kappa)^2) \pi(M_\kappa|\mathbf{x}) - E[\boldsymbol{\theta}_j|\mathbf{x}]^2 \right]$$

Empirical Example

- ▶ Raftery reanalyzes Ehrlich's 1933-1969 state level crime data (a famous and highly criticized study of the economic motivations for crime).
- ▶ The first published instance of a multivariate linear model on deterrence.
- ▶ Outcome variable: crime rate.
- ▶ 15 possible explanatory variables: % young male, south, education, police 1960, police 1969, labor part., sex ratio, population, nonwhites, unemployment 14-24, unemployment 35-39, GDP, inequality, probability of prison, prison time.
- ▶ Number of possible models: 32,768, calculated from $Num = \sum_{r=0}^K \binom{K}{r}$

